TẦNG 1. PRESENTATION LAYER

```
class AppSessionState <<global_state>> {
    + system_initialized: bool
    + ad_closed: bool
    + is_logged_in: bool
    + user_role: String
    + username: String
    + user_obj: UserData
    + payment_step: bool
    + current_page: String
    + selected_movie: String
    + selected_seats: List<String>
    + target_date: String
    + target_time: String
    + generated_ticket_ids: List<String>
    + payment_start_time: double
    + booking_success: bool
    + recent_tickets: List<String>
    + config_slider: List<String>
    + config_list: List<String>

    // Lưu trữ Controllers để giữ trạng thái
    + auth_ctrl: AuthController
    + movie_ctrl: MovieController
    + showtime_ctrl: ShowtimeController
    + room_ctrl: RoomController
    + booking_ctrl: BookingController
    + admin_ctrl: AdminController
}

class MainMenu <<boundary>> {
    + render_login_tab(): void
    + render_register_tab(): void
}

class AdminUI <<page>> {
    + view_dashboard_metrics(): void
    + manage_movie_tab(): void
    + manage_showtime_tab(): void
    + manage_room_tab(): void
    + config_display_tab(): void
    + view_top_revenue_tab(): void
```

```
    + sell_ticket_offline_tab(): void
    + manage_tickets_tab(): void
}

class CustomerUI <<page>> {
    + view_home_page(): void
    + view_booking_page(): void
    + view_history_page(): void
}

// Hàm phụ trợ (ui_components.py)
+ navigate_to(page: String, movie: String = ""): void
+ show_advertisement(): void
+ show_popcorn_effect(): void
+ create_premium_movie_card(col: Any, title: String, genre: String, duration: int, price:
double, img_url: String): void
```

TẦNG 2. BUSINESS LOGIC LAYER

```
class GlobalState <<factory>> {
    + init_global_system(): Tuple<AuthController, MovieController, ShowtimeController,
RoomController, BookingController, AdminController>
}

class AuthController {
    - _io_handler: FileIOHandler
    - _user_table: UserHashTable
    - _current_user: UserData
    - _auth_lock: Lock

    - _generate_user_id(): String
    + login(username: String, pass: String): String
    + register(username: String, pass: String, confirm_password: String): bool
    + logout(): void
    + get_current_user(): UserData
    + is_logged_in(): bool
    + is_admin(): bool
    + get_user_table(): UserHashTable
    + get_all_users(): List<UserData>
}

class AdminController {
```

```
    - _movie_controller: MovieController
    - _booking_controller: BookingController
    - _showtime_controller: ShowtimeController
    - _room_controller: RoomController

    + calculate_revenue(): double
    + count_movies(): int
    + count_tickets(): int
    + get_top_movies_by_revenue(limit: int = 10): List<MovieData>
    + sell_ticket_at_counter(movie: MovieData, showtime: Showtime, row: int, col: int):
bool/String
    + generate_report(): Map<String, Any>
    + get_active_tickets(): List<TicketData>
    + admin_delete_movie(movie_id: String): bool
    + admin_delete_room(room_id: String): bool
}

class MovieController {
    - _io_handler: FileIOHandler
    - _movie_list: MovieLinkedList

    + generate_movie_id(): String
    + add_movie(movie: MovieData): bool
    + update_movie(movie_id: String, title: String, genre: String, duration: int, description:
String, base_price: float, poster_path: String): bool
    + delete_movie(movie_id: String): bool
    + search_by_id(movie_id: String): MovieNode
    + search_by_title(title: String): MovieNode
    + get_movie_list(): MovieLinkedList
    + get_movie_data(): List<MovieData>
    + movie_exists(movie_id: String): bool
    + save_data(): void
    + load_ui_config(): Map<String, List<String>>
    + save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
}

class BookingController {
    - _io_handler: FileIOHandler
    - _showtime_controller: ShowtimeController
    - _movie_controller: MovieController
    - _room_controller: RoomController
    - _ticket_list: TicketLinkedList
    - _booking_lock: Lock
```

```
    + parse_seat_id(seat_id: String): Pair<int, int>
    - _generate_ticket_id(): String
    - _sync_matrix_from_tickets(): void
    - _refresh_booking_data_no_lock(): void
    + process_booking(user: UserData, movie: MovieData, showtime: Showtime, seats:
List<Pair<int,int>>): List<String>
    + confirm_bookings_bulk(ticket_ids: List<String>): bool
    + process_counter_booking(movie: MovieData, showtime: Showtime, row: int, col: int):
bool/String
    + get_booking_history(user_id: String): List<TicketData>
    + find_ticket(ticket_id: String): TicketNode
    + generate_ticket_info(ticket_id: String): String
    + get_ticket_list(): TicketLinkedList
    + get_ticket_data(): List<TicketData>
    + cleanup_unfinished_reservations(timeout_minutes: int = 5): void
    + admin_cancel_ticket(ticket_id: String): bool
    + refresh_booking_data(): void
}

class ShowtimeController {
    - _io_handler: FileIOHandler
    - _movie_controller: MovieController
    - _showtime_list: ShowtimeLinkedList

    + generate_showtime_id(): String
    + add_showtime(st: Showtime): bool
    + update_showtime(showtime_id: String, new_start_time: String): bool
    + delete_showtime(showtime_id: String, ticket_controller: Any = None): bool
    + find_showtime(showtime_id: String): ShowtimeNode
    + check_seat_status(showtime_id: String, row: int, col: int): bool
    + book_seat(showtime_id: String, row: int, col: int): bool
    + release_seat(showtime_id: String, row: int, col: int): bool
    + get_showtime_list(): ShowtimeLinkedList
    + get_showtime_data(): List<Showtime>
    + change_seat_status_by_admin(showtime_id: String, row: int, col: int, new_status: int):
bool
    + get_schedule_by_date(target_date: String): List<List<Any>>
    + extract_date(showtime: Showtime): String
    + extract_time(showtime: Showtime): String
    + get_unique_sorted_dates(showtimes_list: List<Showtime>): List<String>
    + get_unique_sorted_times(showtimes_list: List<Showtime>): List<String>
    + get_showtimes_by_movie(movie_id: String): List<Showtime>
    + find_exact_showtime(movie_id: String, target_date: String, target_time: String):
Showtime
```

```
}

class RoomController {
   - _io_handler: FileIOHandler
   - _room_list: RoomLinkedList

   + add_room(room: Room): bool
   + update_room(room_id: String, new_name: String): bool
   + delete_room(room_id: String): bool
   + find_room(room_id: String): RoomNode
   + find_room_by_name(room_name: String): Room
   + get_room_list(): RoomLinkedList
   + get_room_data(): List<Room>
   + count_rooms(): int
}
```

TẦNG 3. DATA STRUCTURE LAYER

```
class Node {
   + data: Any
   + next: Node
   + get_data(): Any
   + set_data(data: Any): void
   + get_next(): Node
   + set_next(next_node: Node): void
}

class UserNode <<node>> {}          // Các hàm thực thể kế thừa hàm Node cha
class TicketNode <<node>> {}
class MovieNode <<node>> {}
class ShowtimeNode <<node>> {}
class RoomNode <<node>> {}

class Array2D {
   + rows: int
   + cols: int
   + data: List<Any>
   + get_val(r: int, c: int): Any
   + set_val(r: int, c: int, value: Any): void
}

class UserHashTable {
```

```
    - _SIZE: int
    - _table: List<UserNode>
    + hash_function(key: String): int
    + insert(username: String, value: UserData): void
    + get(username: String): UserData
    + contains(username: String): bool
    + remove(username: String): bool
    + display(): void
    + get_table(): List<UserNode>
    + get_all(): List<UserData>
}

class TicketLinkedList {
    - _head: TicketNode
    - _tail: TicketNode
    + add_ticket(ticket: TicketData): void
    + find_ticket(ticket_id: String): TicketNode
    + remove_ticket(ticket_id: String): bool
    + find_by_user(user_id: String): List<TicketData>
    + traverse(): void
    + get_head(): TicketNode
    + get_tail(): TicketNode
    + clear(): void
}

class MovieLinkedList {
    - _head: MovieNode
    - _tail: MovieNode
    + add_movie(movie: MovieData): void
    + search_movie(title: String): MovieNode
    + search_id(movie_id: String): MovieNode
    + remove_movie(movie_id: String): bool
    + sort_by_revenue_logic(): void
    + traverse(): void
    + get_head(): MovieNode
}

class ShowtimeLinkedList {
    - _head: ShowtimeNode
    - _tail: ShowtimeNode
    + add_showtime(st: Showtime): void
    + find_showtime(showtime_id: String): ShowtimeNode
    + find_by_movie(movie_id: String): List<Showtime>
    + remove_showtime(showtime_id: String): bool
```

```
    + traverse(): void
    + get_head(): ShowtimeNode
}

class RoomLinkedList {
    - _head: RoomNode
    - _tail: RoomNode
    + add_room(room: Room): void
    + find_room(room_id: String): RoomNode
    + remove_room(room_id: String): bool
    + traverse(): void
    + get_head(): RoomNode
}

// Mối quan hệ kế thừa
UserNode --> Node
TicketNode --> Node
MovieNode --> Node
ShowtimeNode --> Node
RoomNode --> Node
```

TẦNG 4. DATA MODEL LAYER

```
class SeatStatus <<enum>> {
    + EMPTY: int = 0
    + RESERVED: int = 1
    + BOOKED: int = 2
    + BLOCKED: int = -1
}

class UserData <<entity>> {
    - _username: String
    - _password: String
    - _role: String
    - _user_id: String

    + get_username(): String
    + get_password(): String
    + get_role(): String
    + get_user_id(): String
    + check_password(password: String): bool
}
```

```
class TicketData <<entity>> {
    - _ticket_id: String
    - _user_id: String
    - _movie_id: String
    - _seat_id: String
    - _status: String
    - _showtime_id: String
    - _room_id: String
    - _price: double
    - _booking_time: String

    + get_ticket_id(): String
    + get_user_id(): String
    + get_movie_id(): String
    + get_seat_id(): String
    + get_status(): String
    + get_showtime_id(): String
    + get_room_id(): String
    + get_price(): double
    + get_booking_time(): String
    + set_status(status: String): void
}

class MovieData <<entity>> {
    - _movie_id: String
    - _title: String
    - _genre: String
    - _duration: int
    - _description: String
    - _base_price: double
    - _poster_path: String
    - _revenue: double

    + get_movie_id(): String
    + get_title(): String
    + get_genre(): String
    + get_duration(): int
    + get_description(): String
    + get_base_price(): double
    + get_poster_path(): String
    + get_revenue(): double

    + set_title(title: String): void
    + set_genre(genre: String): void
```

```
    + set_duration(duration: int): void
    + set_description(description: String): void
    + set_base_price(base_price: double): void
    + set_poster_path(poster_path: String): void
    + set_revenue(revenue: double): void
    + add_revenue(amount: double): void
}

class SeatMatrix {
    - _rows: int
    - _cols: int
    - _seats: Array2D

    + get_rows(): int
    + get_cols(): int
    + generate_seat_id(r: int, c: int): String
    + check_status(r: int, c: int): int
    + reserve_seat(r: int, c: int): bool
    + book_seat(r: int, c: int): bool
    + release_seat(r: int, c: int): void
    + get_seats(): Array2D
    + load_matrix(seats_array: List<List<int>>): void
    + set_seat_status(r: int, c: int, status: int): bool
    + reset_all(): void
}

class Room <<entity>> {
    - _room_id: String
    - _room_name: String
    - _rows: int
    - _cols: int
    - _capacity: int

    + get_room_id(): String
    + get_room_name(): String
    + get_rows(): int
    + get_cols(): int
    + get_capacity(): int
    + set_room_name(new_name: String): void
}

class Showtime <<entity>> {
    - _showtime_id: String
    - _movie_id: String
```

```
    - _start_time: String
    - _room_id: String
    - _seat_matrix: SeatMatrix

    + get_showtime_id(): String
    + get_movie_id(): String
    + get_start_time(): String
    + get_room_id(): String
    + get_seat_matrix(): SeatMatrix
    + set_start_time(new_time: String): void
    + get_available_seats(): int
}

class FileIOHandler {
    + base_path: String
    + users_file: String
    + movies_file: String
    + rooms_file: String
    + showtimes_file: String
    + tickets_file: String

    + load_users(table: UserHashTable): void
    + save_users(table: UserHashTable): void
    + load_ui_config(): Map<String, List<String>>
    + save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
    + load_movies(movie_list: MovieLinkedList): void
    + save_movies(movie_list: MovieLinkedList): void
    + load_rooms(room_list: RoomLinkedList): void
    + save_rooms(room_list: RoomLinkedList): void
    + load_showtimes(showtime_list: ShowtimeLinkedList): void
    + save_showtimes(showtime_list: ShowtimeLinkedList): void
    + load_tickets(ticket_list: TicketLinkedList): void
    + save_tickets(ticket_list: TicketLinkedList): void
    + loadData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
    + saveData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
}
```